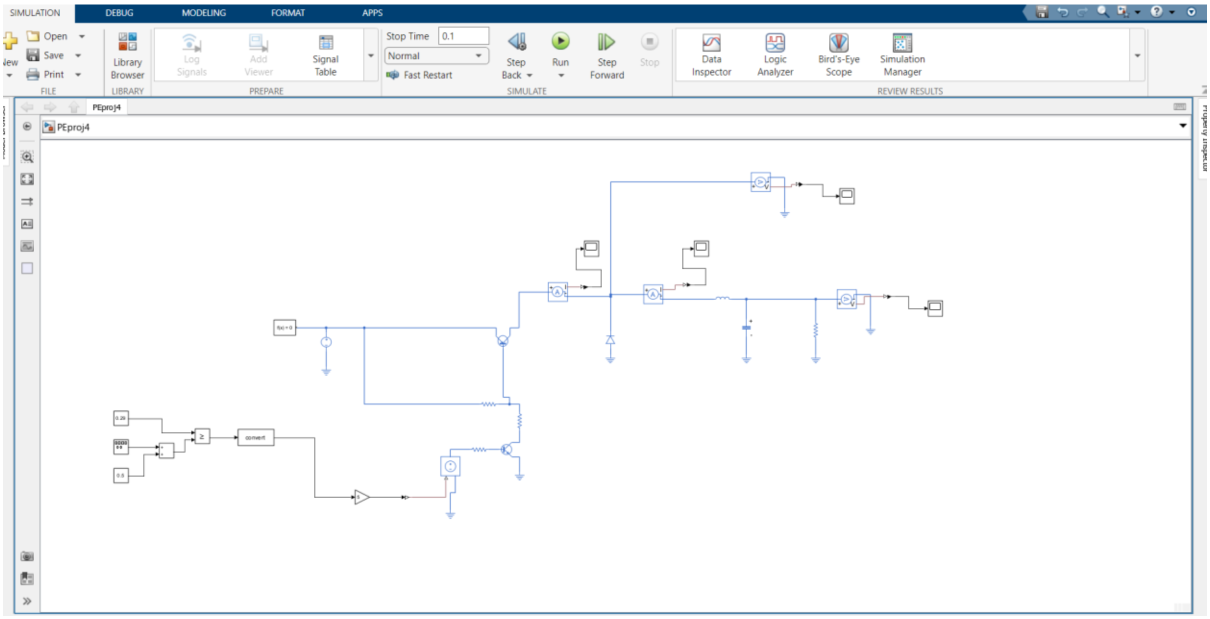
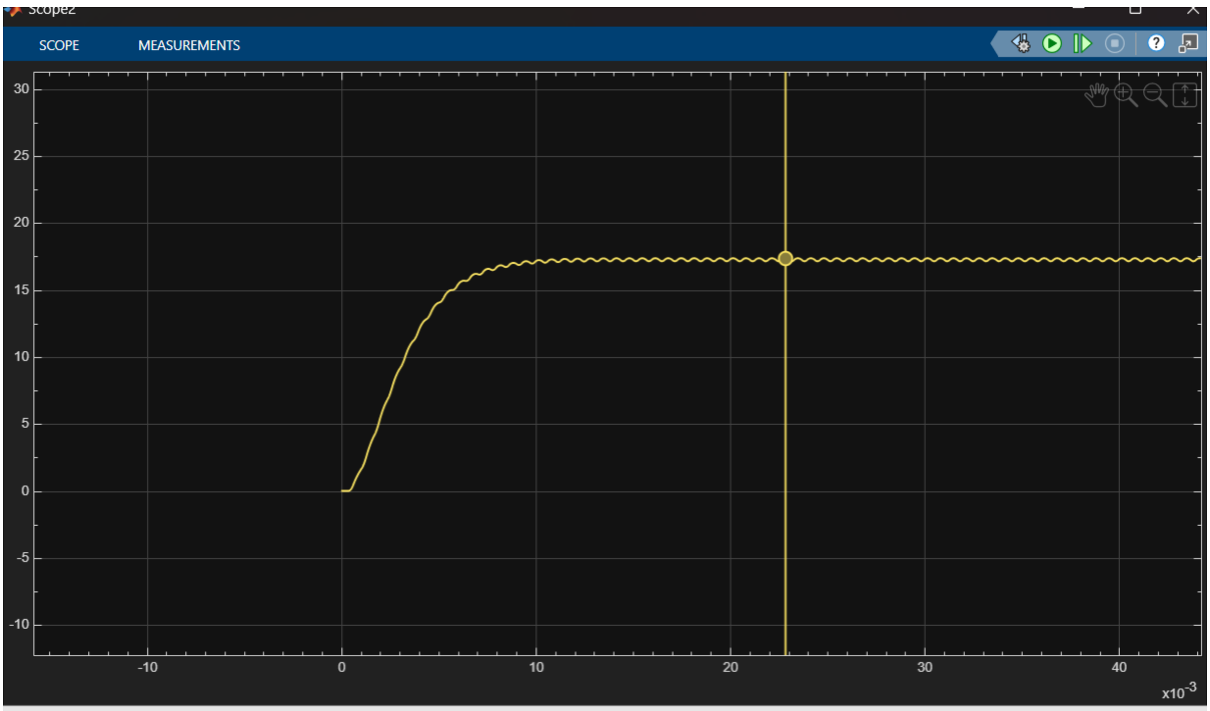
PROJECT PART 4 PE



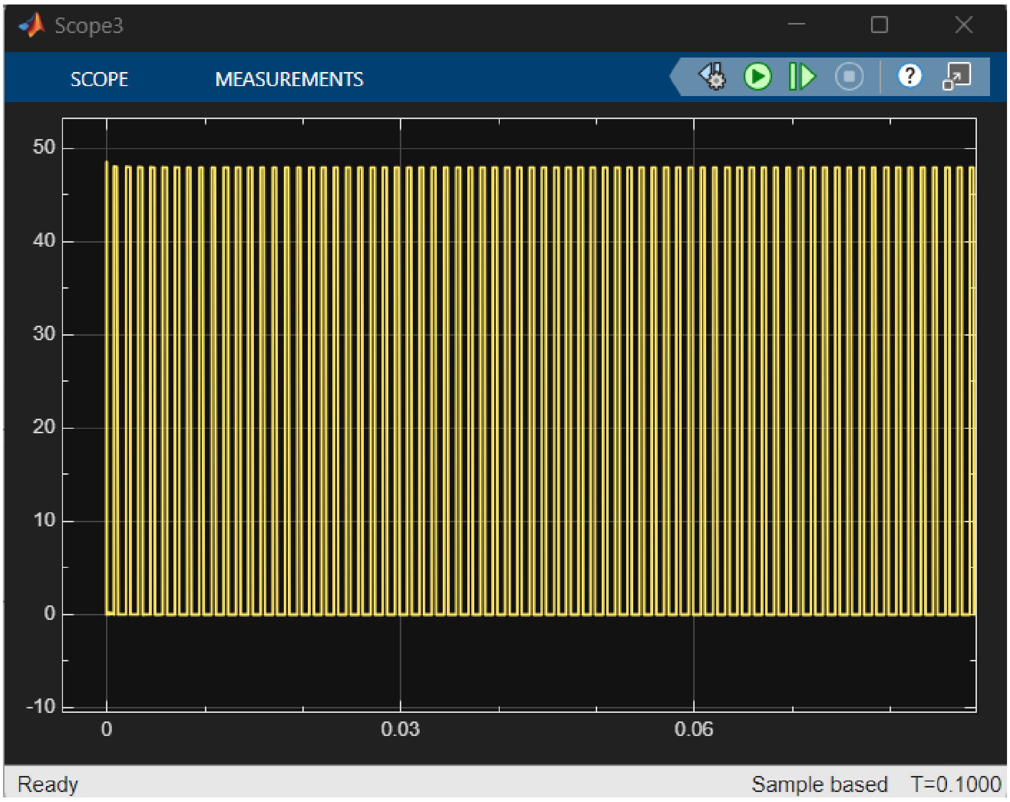
a) IDEAL SIMULATION:

(components without residual elements)

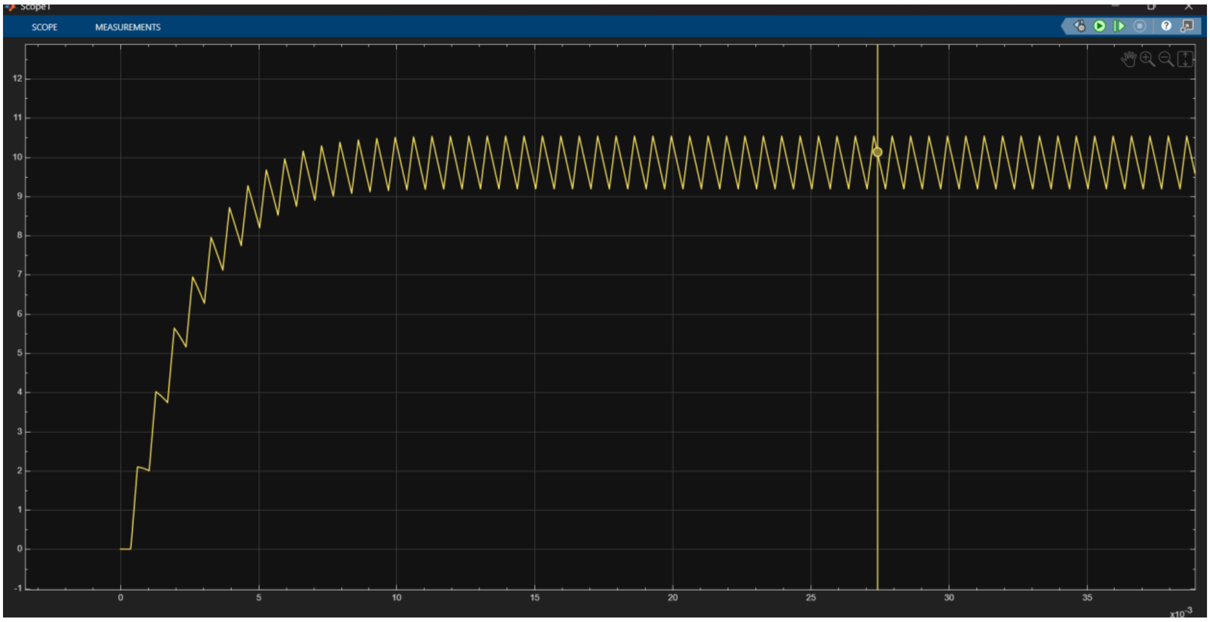
Sensor1:



Sensor2:



The output rises to around 17,6 V with minimal ripple because all components are ideal. The ripple appears because of the normal switching behavior.

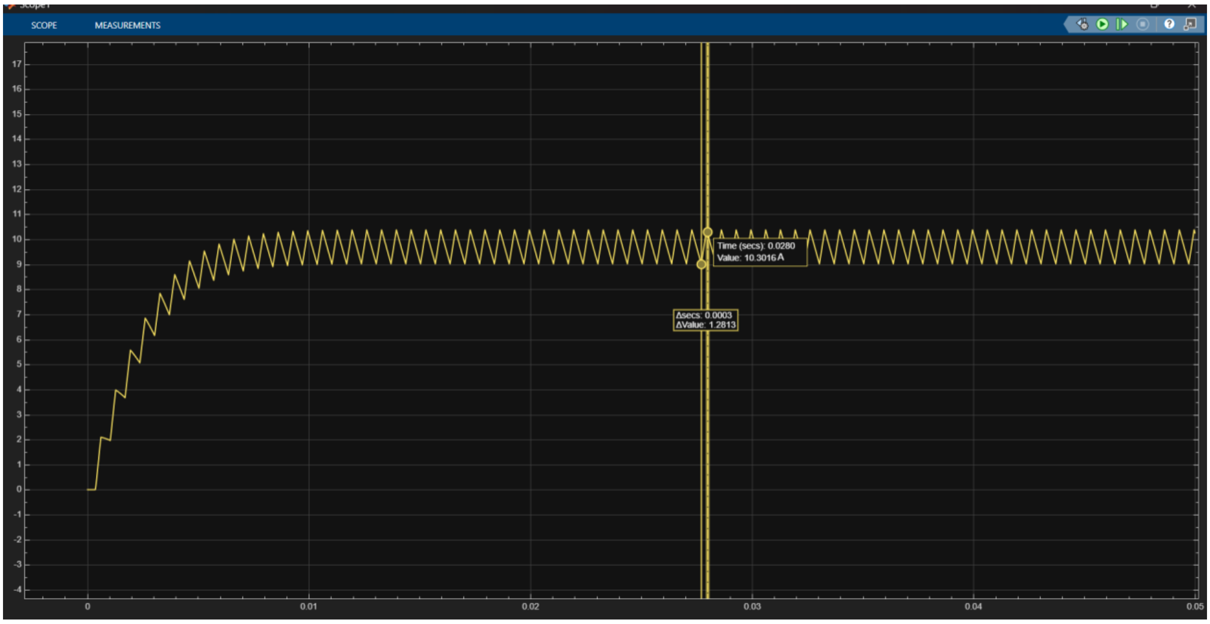


The inductor current shows the triangular waveform with an average value of 9.7A and peak-to-peak ripple of 1.5A..

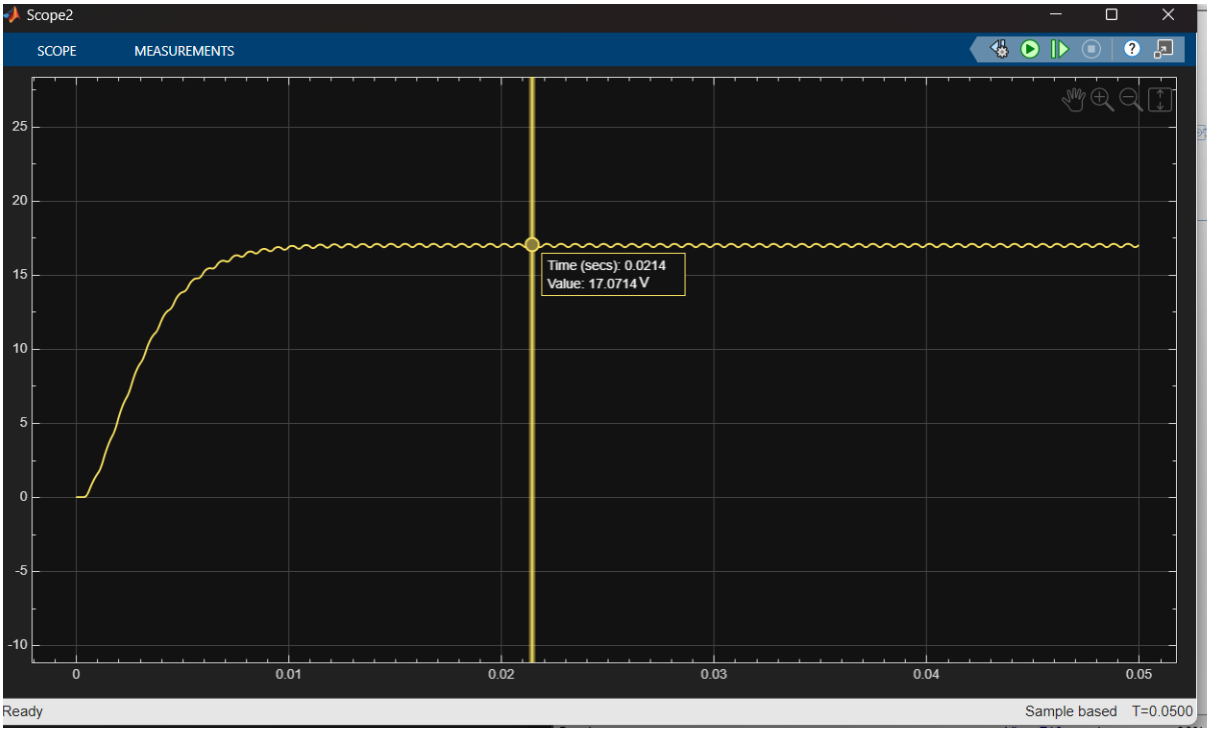
b) SIMULATION WITH RESIDUAL ELEMENTS

1. Diode changed (Forward voltage: **0.55 V** instead of 0.1V)

Current:



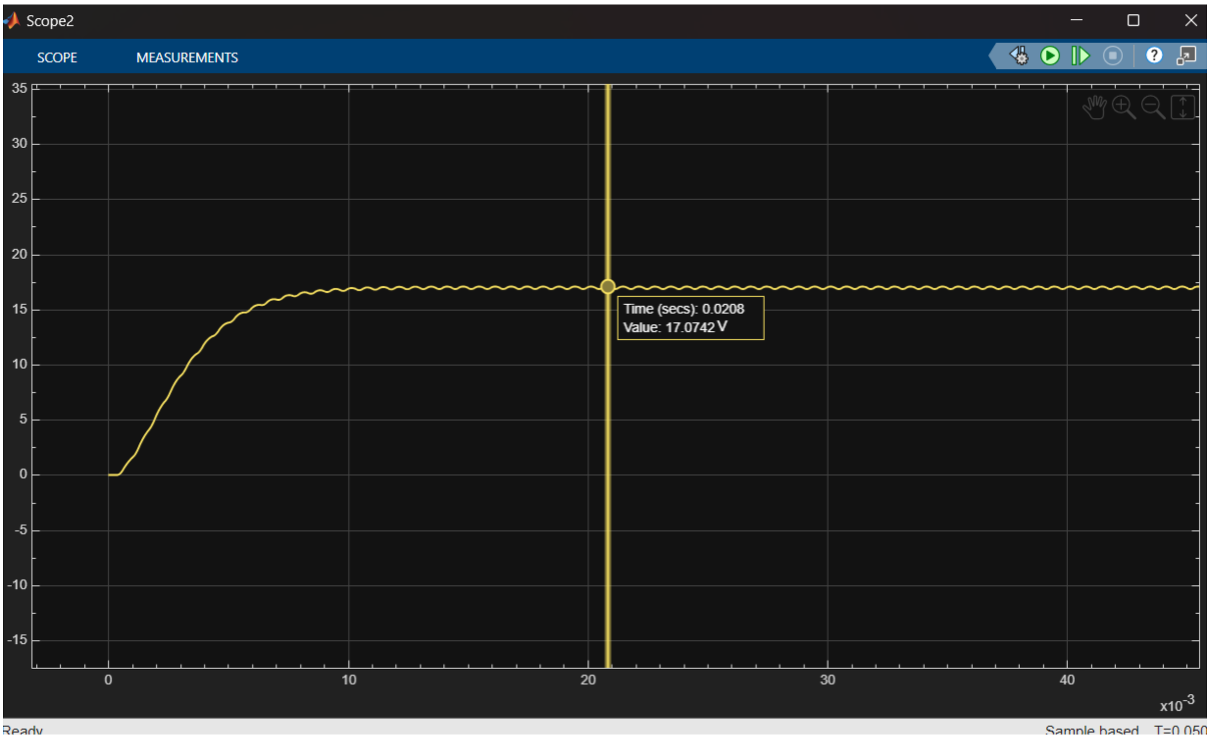
Voltage:

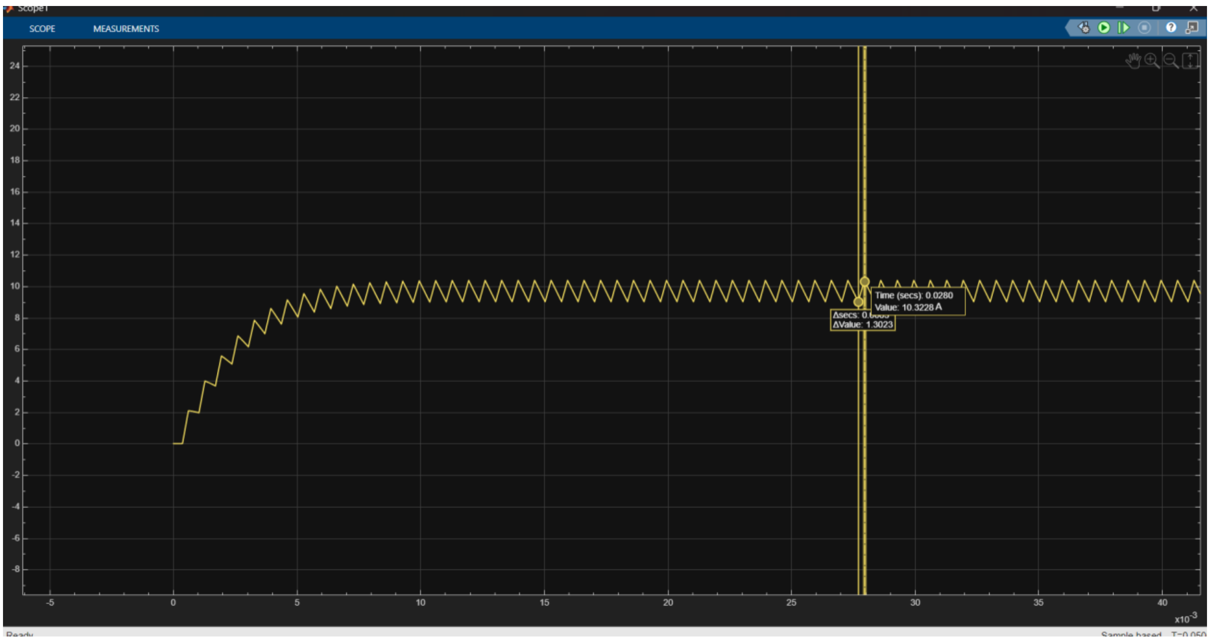


Output voltage decreased slightly (a bit more voltage is lost in the diode)

Current increased since to maintain the output power the circuit draws more current

2. Capacitor: series resistance ESR 0.022 Ω

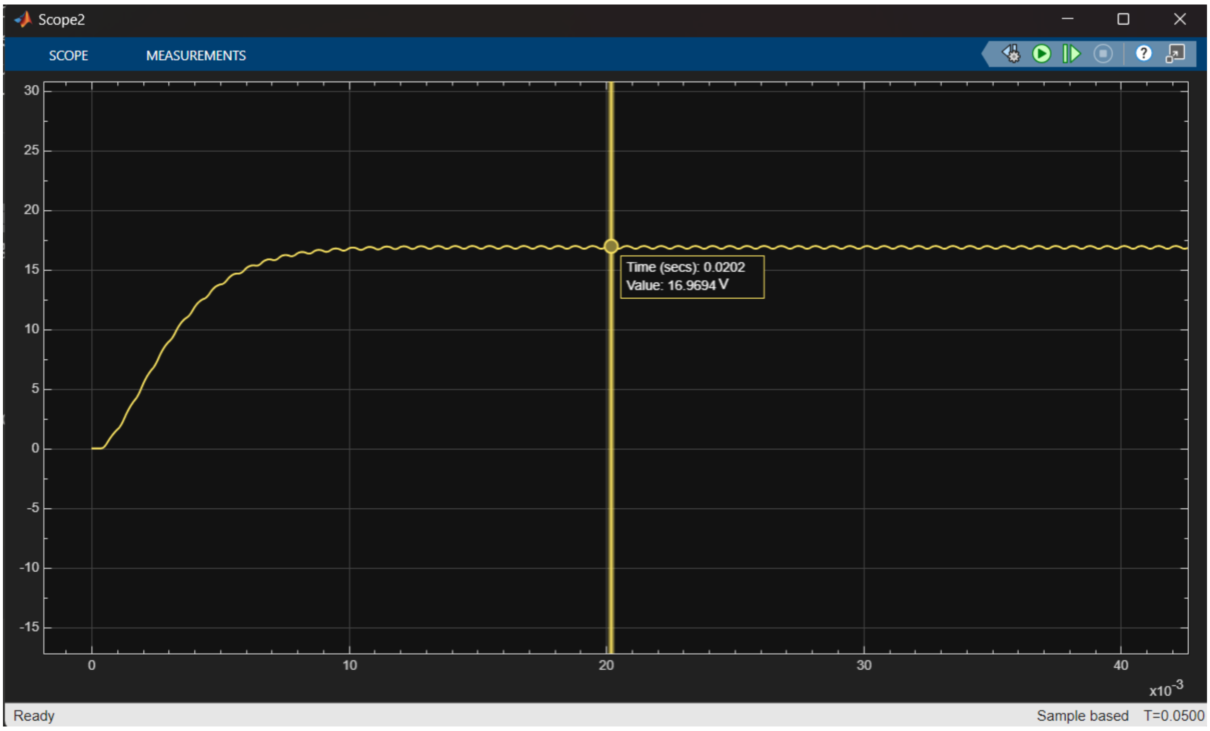


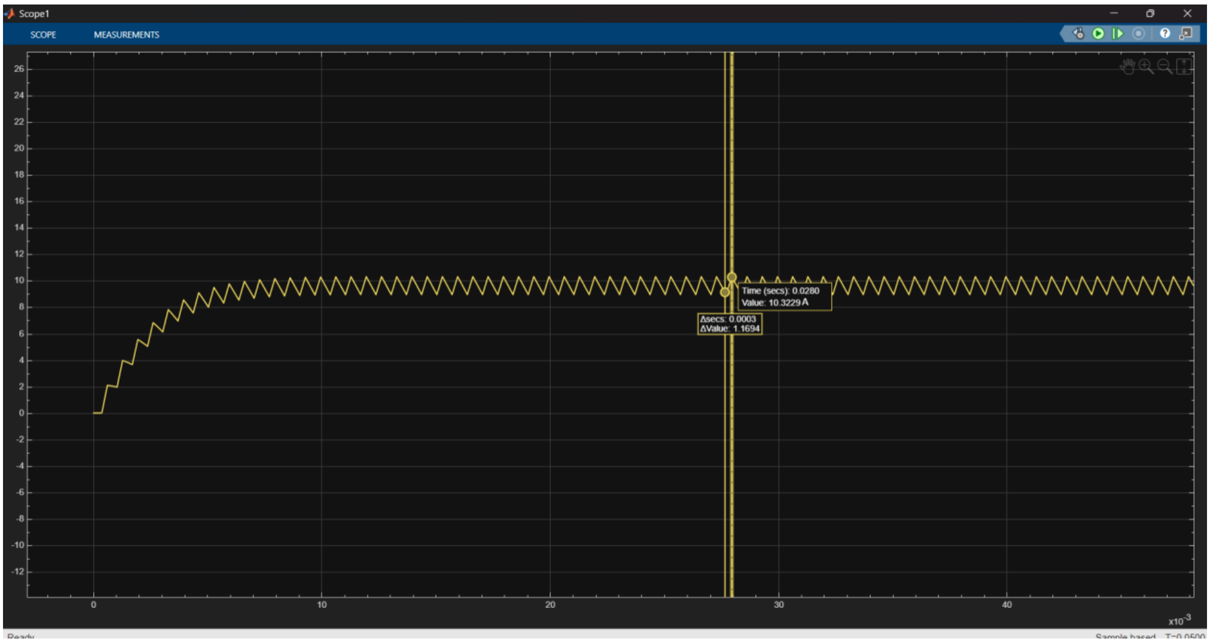


Current remained the same, just a bit increased.

The output voltage has a small but still visible ripple.

3. Series resistance: DC resistance = 0.0103 Ω

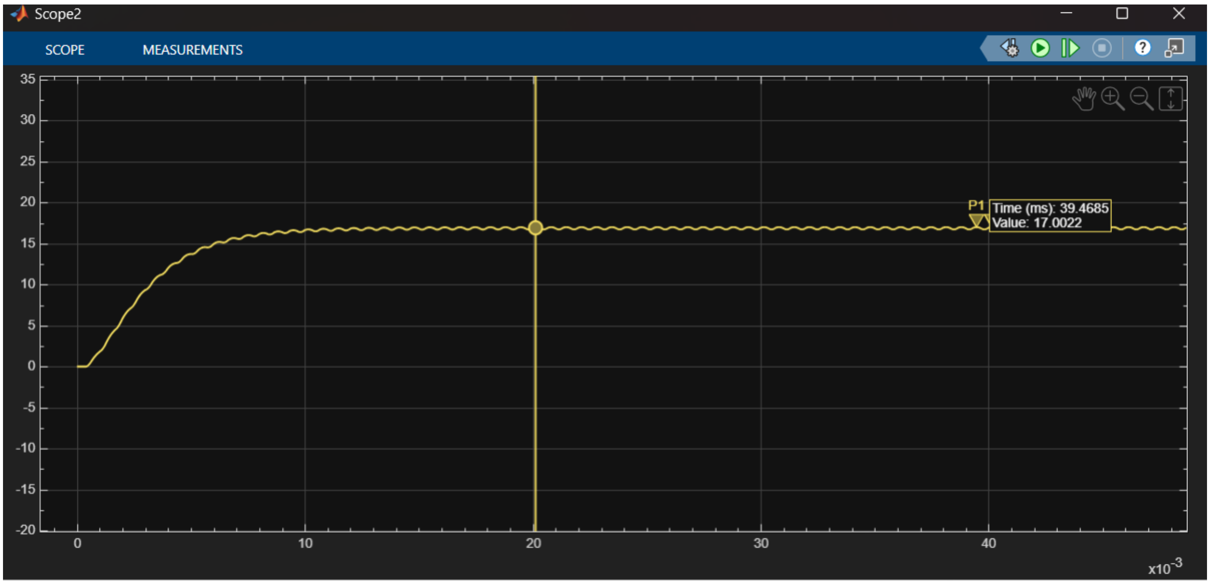




Output voltage decreased because DCR causes a voltage drop.

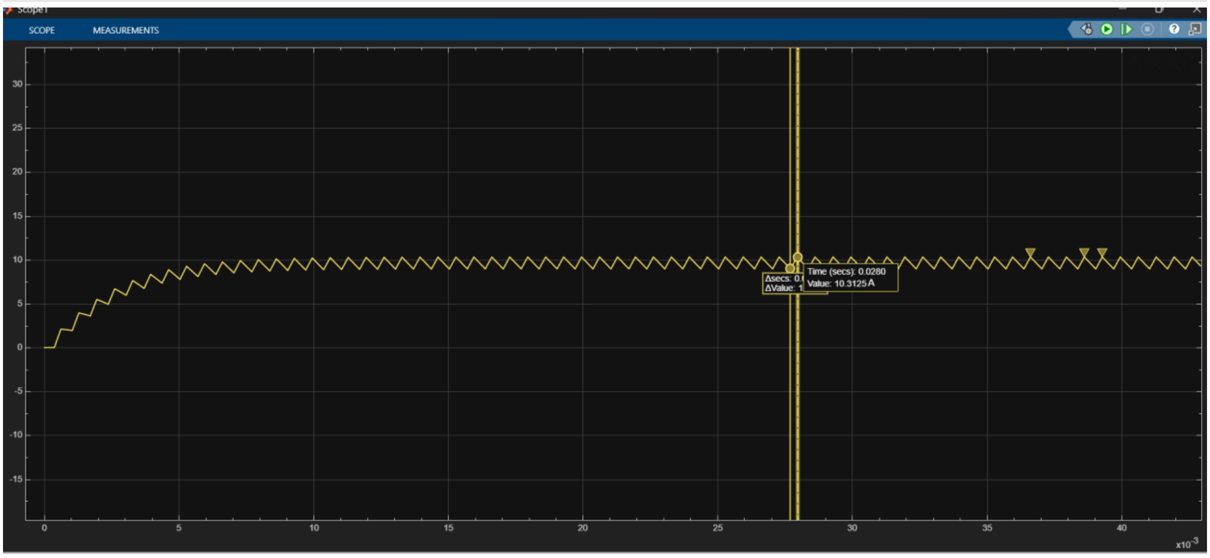
Current remained kind of the same just the proximity (ripple) is decreased slightly.

4. Capacitor value to minimum tolerance value: C = 400 μF (from 500 μF)



Peaks

PS-Simulink Converter2	P1	P2	P3
Time (ms)	39.4685	40.1355	41.4671
Value	17.0022	17.0021	17.0021

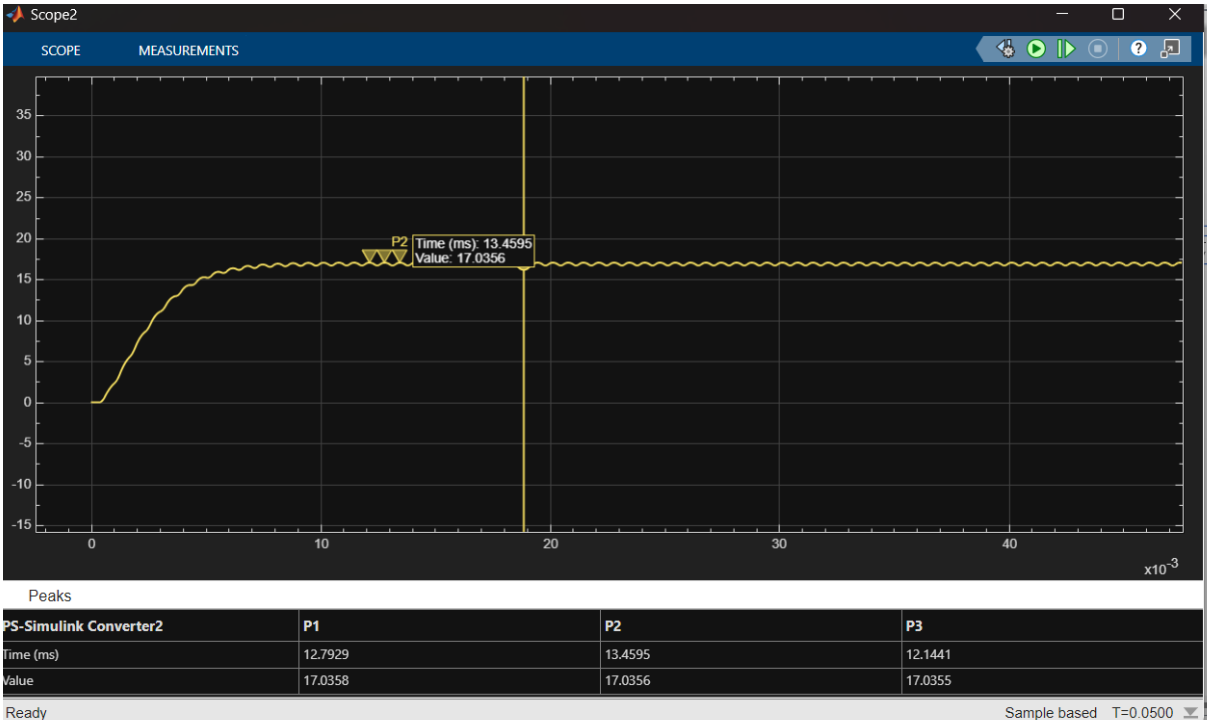


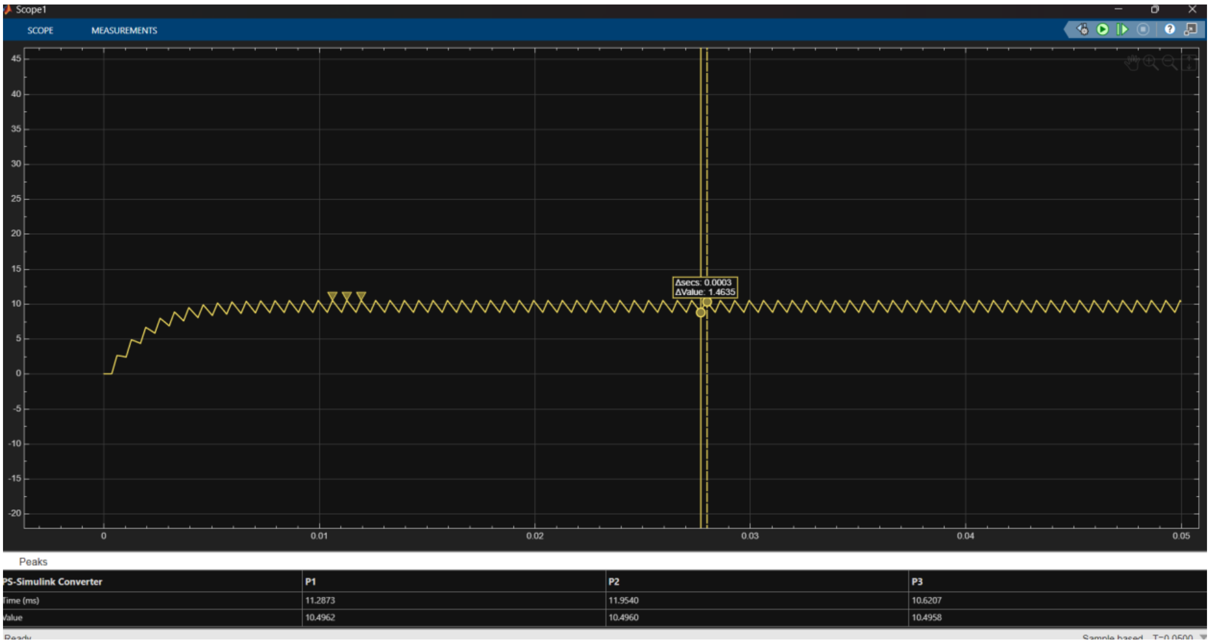
Peaks

PS-Simulink Converter	P1	P2	P3
Time (ms)	36.6207	38.6207	39.2873
Value	[illegible]	[illegible]	[illegible]

The ripple is a bit larger at output voltage (a smaller capacitor doesnt have the ability to make a smoother voltage ripple)

5. Inductor: Tolerance application: Apply minimum tolerance value L = 4.403 mH (from 5.504 mH) - 5.504mH × 0.8





For current, the ripple increased a lot because the inductor has less ability to oppose current changes.

Output voltage also increased a little.